MOLLY OHSE

PhD Candidate in Wildlife Ecotoxicology & Ecophysiology
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PROFILE

Ornithologist and ecotoxicologist studying how mercury and other metals shape the physiology, life history and fitness of wild birds. My work links long-term, individual-based field data from a white-throated dipper (*Cinclus cinclus*) population with lab measures of mitochondrial function, oxidative stress, and corticosterone, analysed with Bayesian mixed models, quantitative genetics, and phylogenetic meta-analysis.

EDUCATION

Swiss Ornithological Institute & University of Lausanne Oct 2023 – Oct 2027 (expected)
PhD in Life Sciences *Sempach & Lausanne, Switzerland*

- Thesis: *Effects of chemical pollutants on mitochondrial traits, individual fitness and trait heritability in wild birds: development of generic tools and a case study of mercury effects in a natural population of white-throated dippers* (SNSF project grant 212971)
- Supervisors: Dr Pierre Bize (Vogelwarte), Prof. Philippe Christe (UNIL)
- Collaborators: Dr Blandine Doligez (CNRS Lyon), Dr Antoine Stier (CNRS, IPHC Strasbourg)

University of Aberdeen Oct 2021 – Dec 2022
MSc in Biology (Research), with Distinction *Aberdeen, UK*

- Thesis: *Sources of variation and consequences of exposure to metals and metalloids in a white-throated dipper population in France*
- Supervisors: Dr Gareth Norton, Dr Pierre Bize, Dr Blandine Doligez

University of Aberdeen Sept 2017 – July 2021
BSc (Hons) Conservation Biology, Upper Second Class *Aberdeen, UK*

- Thesis: *Sources of variation in DDT exposure and consequences for reproduction in wild birds: a phylogenetically controlled meta-analysis*
- Supervisor: Dr Pierre Bize · Accredited by the Royal Society of Biology
- Erasmus+ exchange semester, University of Tromsø, Norway (Jan – June 2019)

RESEARCH EXPERIENCE

Doctoral Researcher Oct 2023 – present
Swiss Ornithological Institute & University of Lausanne *Switzerland*

- Led a two-year partial cross-fostering experiment (62 nest pairs, 459 nestlings) separating maternal from post-hatching mercury exposure in a wild population
- Co-developed a plate-reader respirometry assay for mitochondrial function in avian red blood cells, validated against high-resolution respirometry
- Co-leading a phylogenetic meta-analysis of metal and metalloid effects on oxidative stress in birds (55 studies, 609 effect sizes)
- Manage the master database and social pedigree of the long-term Chartreuse dipper population (thousands of individuals, ~600 nests over nearly a decade)
- Coordinate fieldwork with student teams on remote mountain rivers: nestling ringing, biometrics, sampling of blood, feathers and preen oil, nest monitoring and video recording

Research Assistant Feb 2022 – Oct 2023
Swiss Ornithological Institute *Sempach, Switzerland*

- Updated the DDT/DDE meta-analysis and prepared MSc thesis work for publication
- Ran mercury analyses of dipper feathers (Direct Mercury Analyser)

- Led fieldwork and weekly nest monitoring for a long-term dipper population in the French Alps

PUBLICATIONS

Published

[1] Hadjadji C.†, **Ohse M.**†, Stier A. & Bize P. (2026). High-throughput mitochondrial aerobic metabolism measurement: assessing a plate-reader approach using avian blood cells. *Biology Open*. PMID 42803112.

Submitted / in revision

[2] **Ohse M.**, Hinninger L., Norton G., Doligez B. & Bize P. Repeated measures of feather mercury in an aquatic-feeding songbird reveal persistent territory-associated variation and consistent individual differences. *Environmental Monitoring and Assessment* (revision submitted Nov 2026). Preprint on Research Square.

[3] Hinninger L., **Ohse M.**, Doligez B. & Bize P. Spatial scale and site fidelity shape the correlation between blood and feather mercury in a riverine bird. *Ecological Indicators* (submitted).

In preparation

[4] **Ohse M.**† & Hadjadji C.†, Bize P. Context-dependent effects of metals and metalloids on avian oxidative stress: a phylogenetic meta-analysis of experimental and field studies. Target: *Biological Reviews*.

[5] **Ohse M.**, Gray F., Nsude C. & Bize P. Estimating chemical risk thresholds in the wild: a phylogenetically controlled meta-analysis and dose-response modelling of DDE and reproduction in birds. Target: *Chemosphere*.

[6] **Ohse M.**, Cillard A., Bastianelli M., Doligez B. & Bize P. From exposure to phenotype: evaluating mitochondrial and oxidative stress pathways linking mercury to nestling growth in a riverine passerine. Target: *Functional Ecology*.

[7] **Ohse M.**, Norton G., Cochrane N., Doligez B. & Bize P. Repeatability of feather metal and metalloid concentrations in an aquatic-feeding songbird: landscape effects and associations with phenotype and reproduction. Target: *Archives of Environmental Contamination and Toxicology*.

[8] **Ohse M.**, Doligez B. & Bize P. Linking river and landscape indicators to occupancy and breeding performance in white-throated dippers.

[9] **Ohse M.**, Hadjadji C., Masoero G., Doligez B. & Bize P. Small-volume blood sampling is not associated with reduced fitness in nestling White-throated Dippers (*Cinclus cinclus*) and Alpine Swifts (*Tachymarptis melba*).

† equal contribution

GRANTS

Swiss Society of Toxicology travel grant 2026
International Ornithological Congress, Mérida

SUPERVISION

Chloé Maret 2026 – June 2027
MSc, University of Lausanne (with Dr Pierre Bize)
Heritability of mercury contamination and its effects on biometric traits in white-throated dippers

Lara Hinninger 2025
MSc, University of Zurich (with Dr Pierre Bize)
Cascading effects of mercury on oxidative stress and individual fitness in white-throated dippers

SERVICE & PROFESSIONAL ACTIVITIES

- Session organiser, Society for Experimental Biology (SEB) Annual Conference, Glasgow, 2027

- Symposium co-convener, *Avian mitochondria: Powering ecology, evolution, and conservation*, 15th European Ornithologists' Union (EOU) Conference, Bangor, UK, 2025 (with C. Hadjadji & P. Bize)
- Member, Swiss Society of Toxicology (SST)

Skills

Laboratory	Direct Mercury Analyser; ICP-MS (acid and microwave digestion of feathers); plate-reader respirometry (VANTAstar); oxidative stress assays; feather corticosterone
Field	Nestling ringing; handling of adult and nestling birds; biometrics; blood, feather and preen-oil sampling; nest monitoring and video analysis; cross-fostering; team coordination
Statistics (R)	Bayesian mixed models (MCMCglmm); multilevel and phylogenetic meta-analysis (metafor); repeatability; quantitative-genetic animal models; SEM; PCA; PRISMA-EcoEvo
Data	Long-term individual-based databases and pedigrees; R Markdown; open data repositories; QGIS
Languages	English (native); Italian (B1)

Outreach, Education & Animal Care

Natural Areas Instructor, *Downers Grove Park District, IL, USA* June – Sept 2021
Led nature programmes for children and families; beekeeping and reptile care

Animal Care Intern, *Fox Valley Wildlife Center, Elburn, IL, USA* June – Sept 2019
Feeding, medication and treatment of injured wildlife; triage of animals brought in by the public

Elephant Care Volunteer, *Wildlife Friends Foundation Thailand* Dec 2015 – Jan 2016
Led a daily volunteer team caring for a rescued elephant

Camp Counsellor, *YMCA Camp Wapsie, Coggon, IA, USA* May – Aug 2015
Supervised ~150 children (aged 5–16) weekly at a residential camp

Guest Engagement Work-Study, *Shedd Aquarium, Chicago, IL, USA* Feb – May 2015
Conservation interpretation and public presentations

Kennel Volunteer, *West Suburban Humane Society, Downers Grove, IL, USA* Mar – Dec 2015

References

Available on request.